Business Problem Statement

Business Problem

Swire Coca-Cola (Swire) recently launched MyCoke360, a B2B platform for Foodservice On-Premise (FSOP) customers to purchase Coca-Cola products. Swire is incurring losses each order cycle due to cart abandonment, where customers add products to their carts but do not submit orders by the scheduled date. This problem disrupts plant logistics, reduces revenue, and strains customer relationships.

Benefit of a Solution

The purpose of the proposed solution is to prevent cart abandonment, or at the very least, understand why it happens. This project will provide insights into what events or sequence of events correlate with cart abandonment, what causes a customer to return to a previously abandoned cart, what happens after a cart is abandoned, what products appear most frequently in abandoned carts, and what the overall financial impacts of cart abandonment are to Swire Coca-Cola.

These insights will benefit Swire Coca-Cola by enabling them to quantify (by product and revenue), predict, and prevent cart abandonment with the potential to optimize the MyCoke360 platform.

Success Metrics

Quantifying lost revenue and mix distortion will enable the business to size the opportunity, prioritize fixes, and target interventions where they matter most. Clear measurement of lost dollars, recovered dollars, and net revenue impact will provide an objective basis for investment decisions. Identifying which brands and pack types are overrepresented in abandoned value versus realized sales will inform pricing, merchandising, and inventory actions that protect revenue while improving the customer experience.

Success also requires an operational definition of “cart abandonment” and attribution of outcomes. For example, whether a customer who abandons a cart later purchases through a different channel or in a subsequent order cycle. Accordingly, orders should be linked and analyzed at a granular level by customer, brand, pack, and cycle to distinguish truly abandoned carts from postponed purchases.

Analytics Approach

This will be a descriptive analytics project. We will begin with an exploratory data analysis taking information from Swire's Google Analytics platform, and other internal sources. The data will include webpage interactions, order information, customer information, product information, shipping information, and sales information.

After the data has been cleaned and normalized, we will identify the best definition for cart abandonment, taking into account customer patterns and business needs. We will then identify useful predictors, highly-correlated variables, and other data trends that could help understand cart abandonment, including the financial impact. Machine learning model(s) will be developed to see how accurately we can explain cart abandonment.

Scope

The project will deliver an exploratory data analysis (eda), a notebook of implementable models, and a presentation on the project's insights. The eda will serve to establish the existing extent of cart abandonment. The notebook of implementable model(s) will serve two purposes. First, as a template for Swire Coca-cola's data science department to implement any solutions they find useful. Second, as a proof to defend accuracy of the insights. The presentation will serve to communicate the insights discovered, success metrics, and recommendations.

Additional deliverables such as website changes, or cross-channel churn analysis will need to be agreed upon with the analyst and added to a revised business problem statement.

Details

Project Team: Ali Ladha, Robert Stohel, Cyrus Sobhani, and Sterling Leduc will serve as data scientists.

This timeline should suffice for the completion of the project by December 3, 2025:
- Business problem statement finalized and submitted by September 14, 2025.
- Exploratory data analysis and data cleaning completed by October 4, 2025.
- Prototype model delivered by October 18, 2025.
- Model selection finalized by October 25, 2025.
- Practice presentation recording submitted by November 1, 2025.
- Final presentation revisions completed by November 8, 2025.
- Final presentation slides submitted by November 15, 2025.
- Final stakeholder presentation delivered on November 16, 2025.